cells and have kidney shaped nucleus. They constitute 1-3% of the total WBCs. The macrophages of the central nervous system are the 'microglia', in the sinusoids of the liver they are called **'Kupffer cells'** and in the pulmonary region they are the **'alveolar macrophages'**.

**Platelets** are also called **thrombocytes** that are produced from megakaryocytes (special cells in bone marrow) and lack nuclei. Blood normally contains 1, 50,000 -3, 50,000 platelets $mm^{-3}$ of blood. They secrete substances involved in coagulation or clotting of blood. The reduction in platelet number can lead to clotting disorders that result in excessive loss of blood from the body.

### 7.1.3 Blood groups

Commonly two types of blood groupings are done. They are ABO and Rh which are widely used all over the world.

### ABO blood grouping

Depending on the presence or absence of surface antigens on the RBCs, blood group in individual belongs to four different types namely, A, B, AB and O. The plasma of A, B and O individuals have natural antibodies (agglutinins) in them. Surface antigens are called **agglutinogens**. The antibodies **(agglutinin)** acting on agglutinogen A is called anti A and the agglutinin acting on agglutinogen B is called anti B. Agglutinogens are absent in O blood group. Agglutinogens A and B are present in AB blood group and do not contain anti A and anti B in them. Distribution of antigens and antibodies in blood groups are shown in Table 7.1. A, B and O are major allelic genes in ABO systems. All agglutinogens contain sucrose, D-galactose, N-acetyl glucosamine and 11 terminal amino acids. The attachments of the terminal amino acids are dependent on the gene products of A and B. The reaction is catalysed by glycosyl transferase.

**Table 7.1** Distribution of antigens and antibodies in different blood groups

| Blood group | Agglutinogens (antigens) on the RBC | Agglutinin (antibodies) in the plasma |
|---|---|---|
| A | A | Anti B |
| B | B | Anti A |
| AB | AB | No antibodies |
| O | No antigens | Anti A and Anti B |

**Rh factor** is a protein **(D antigen)** present on the surface of the red blood cells in majority (80%) of humans. This protein is similar to the protein present in Rhesus monkey, hence the term Rh. Individuals who carry the antigen D on the surface of the red blood cells are $Rh^+$ (Rh positive) and the individuals who do not carry antigen D, are $Rh^-$ (Rh negative). Rh factor compatibility is also checked before blood transfusion. When a pregnant women is $Rh^-$ and the foetus is $Rh^+$ incompatibility (mismatch) is observed. During the first pregnancy, the $Rh^-$ antigens of the foetus does not get exposed to the mother's blood as both their blood are separated by placenta. However, small amount of the foetal antigen becomes exposed to the mother's blood during the birth of the first child. The mother's blood starts to synthesize D antibodies. But during subsequent pregnancies the Rh antibodies from the mother ($Rh^-$) enters the foetal circulation and destroys the foetal RBCs. This becomes fatal to the foetus



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